

INSuRE Fall 2025
Midterm Report - 10/31/2025

FPGA Side-Channel Analysis

Investigators:

The University of Texas at Dallas

Nicholas Donovan - ngd210000@utdallas.edu
Brandon Wang - blw190004@utdallas.edu
Timothy Stuard - tbs140030@utdallas.edu
Arden Shao - aws240000@utdallas.edu
Naja-Lee Habboush - nih200000@utdallas.edu
Brian Ricks (Faculty Support) - bwr031000@utdallas.edu

Problem Mentors:

Oak Ridge National Laboratory

Tristen Mullins
John Benson

Table of Contents

Table of Contents	2
1.0 - Project Summary	3
1.1 Top Level Research Project Description/Problem Statement	3
1.2 Project Objectives/Goals	3
1.3 - Investigator Biographies	3
2.0 - Research Plan	5
2.1 - Research Plan Summary	5
2.2 - Deliverables	5
2.3 - Progress to Date	6
3.0 - Summary of Results	11
4.0 - Issues and Challenges	12
5.0 - Bibliography	13
Appendix A	14
A.1 - Acronyms	14
A.2 - Definitions	14

1.0 - Project Summary

1.1 Top Level Research Project Description/Problem Statement

FPGAs are a fundamental hardware building block found in various systems such as aerospace, telecommunications, and critical infrastructure (industrial networks). Being able to verify the integrity and security of FPGAs is crucial to the overall security of the systems on which they are built. Our research aims to provide a reliable means to detect malicious modification to FPGAs using information from the FPGA bitstreams and side-channel analysis (e.g. power measurements).

1.2 Project Objectives/Goals

The primary goal of the project is to provide a reliable means to detect undesired hardware program modification through means such as power, electromagnetic, and/or other side-channel analysis. Our main focus is developing a benign program to be loaded onto an FPGA, developing a modified version of the same program to be loaded onto an FPGA, and comparing measurements obtained via side channels for the two programs to determine if side channel measurements can be used as a means for detecting hardware program modification.

1.3 - Investigator Biographies

Arden Shao graduated with a Bachelor's in Computer Science and is currently studying for a Master's in Computer Science in the cybersecurity track as an SFS scholar. He has taken courses on systems security, network security, digital forensics, and machine learning. As well, he has held an internship with a national laboratory that focused on reverse engineering of hardware and firmware. He has participated in various cybersecurity CTF competitions for the Computer Security Group and attended a DoE conference focusing on OT security. In addition, he is an officer for the Cyber Defense and Response Unit, an organization focusing on teaching blue-teaming skills and participating in cybersecurity defense competitions. His primary skills are in reverse engineering, binary analysis, and digital forensics.

Naja-Lee Habboush holds a Bachelors of Science in Computer Science, and is pursuing a Master's in Computer Science with a focus of Cybersecurity at the University of Texas at Dallas. She has conducted research on Ant Colony Optimization, a swarm-intelligence machine-learning algorithm, under the guidance of Dr. Gity Karami. Professionally, she has interned at NCR Atleos, an ATM company, applying her classroom knowledge of cryptography and security protocols in a real-world environment. Among these, she has experience in exploiting vulnerabilities in IoT devices. Her current interests involve cryptography, quantum computing, and system security, studying these through her coursework.

Nicholas Donovan started as an IT administrator before earning a bachelor's degree in Computer Science from UT Dallas. He is currently pursuing a Master's in Computer Science

with a focus on cybersecurity, as an SFS scholar. Alongside his curriculum, Nick currently works with a national laboratory on the Cyber Operations and Incident Response Team, where he focuses on incident response, forensic triage, technique analysis, analytic development, as well as incident response kit tooling and automation. Outside of his professional career, Nick serves as the president of the Cyber Defense and Response Unit, a nonprofit organization dedicated to blue-teaming. In this role, he leads a team of students in training on digital forensics, incident response, and threat hunting. Additionally, he is the operations officer of Computer Security Group, the largest cybersecurity organization on campus, and has represented both the university and the organizations on several competitive cyber teams.

Timothy Stuard is a second-year master's student in Computer Science focusing on Cybersecurity at the University of Texas at Dallas, with a B.S. in Pure Mathematics from the University of Texas at Austin. His interests include cryptography, cyber-physical systems, and malware analysis, with a mathematically rigorous approach to security engineering. He has worked across data/app security, cloud–edge–IoT security, operating systems, and multiprocessor programming, and recently began prototyping CUDA kernels to study GPU performance. He is now sharpening his focus on practical malware analysis from static/dynamic tooling to behavior profiling and evasion-aware detection.

Brandon Wang is currently in his last semester of his CS Master's degree following the Cybersecurity track as a DoD CSA scholar. His coursework has included exploring network security, secure network protocols and practical implementations of them, binary exploitation and defense with inline monitoring, and digital forensics in NTFS. He has prior experience as a SOC analyst and participant/challenge author for CTF competitions, and has expanded his interests into operating system and endpoint security.

2.0 - Research Plan

2.1 - Research Plan Summary

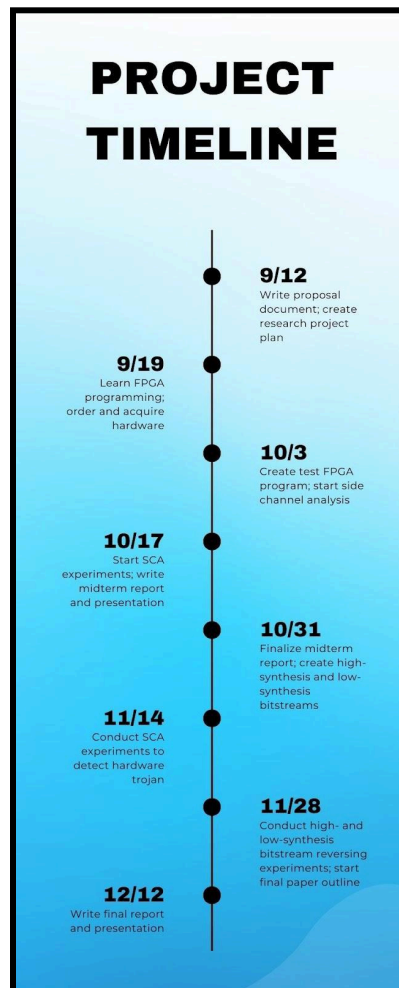
Our overall research project plan is focused on performing experiments on FPGAs in order to determine how to detect hardware trojans and bitstream synthesis procedures. These experiments will be done with the eventual purpose of writing a final report detailing methodology and results, which may potentially be formatted into an academic paper and submitted to a conference if findings are notable. Initially, the team will build a knowledge base by reviewing existing side-channel analysis literature, learning FPGA programming using Vivado and VHDL, and training to use our hardware. Then, we will build upon this knowledge through preliminary efforts to create simple FPGA programs and take power readings using an oscilloscope and electromagnetic readings with a software-defined radio (SDR). Some members of the team will focus on becoming familiar with side-channel analysis, while others will focus on FPGA programming and bitstream synthesis.

The bulk of project time will be spent on designing and conducting side-channel analysis experiments. One FPGA will be unmodified and used as a control, while a hardware trojan will be introduced to a second FPGA for testing purposes. The team will design a test FPGA bitstream, create a modified version that contains a hardware trojan, and place these bitstreams on the aforementioned FPGAs. We will set up experiments to collect data on power and EM emission readings in order to investigate how to detect the hardware trojan through such side-channel analysis. These readings will be used in an attempt to build a classifier based on the family architecture which will detect malicious activity within the FPGA. As well, team members will create high- and low-synthesis versions of a bitstream in order to test effects on subsequent side-channel analysis. If time permits, members will also investigate if the above differences in bitstream synthesis process impacts the information that may be recovered through reverse engineering the bitstream. At the end of the project, the team will put together a final report detailing our experiments and findings.

2.2 - Deliverables

- The deliverables completed so far are as follows:
 - 9/26 - Initial draft of benign traffic light program completed
 - 10/8 - FPGA boards received, along with other side-channel analysis hardware (HackRF and ChipWhisperers)
 - 10/17 - Traffic light program completed and successfully loaded onto an FPGA
 - 10/24 - ChipWhisperer power analysis setup completed
 - 10/29 - Elevator program completed and successfully loaded onto an FPGA
 - 10/31 - Midterm report and presentation, HLS version of program completed and loaded
- The deliverables to be completed are as follows:
 - 11/7 - Completion and loading of a modified version of a benign program

- 11/14 - Power consumption measurements for both the benign program and modified program
- 11/14 - Electromagnetic emission measurements for both programs
- 11/21 - Power and electromagnetic measurements for HLS program and low-level synthesis program
- 11/28 - A report discussing the relationship between the two measurements, most likely within our final report
- For comparison, here is the project timeline used in our proposal:

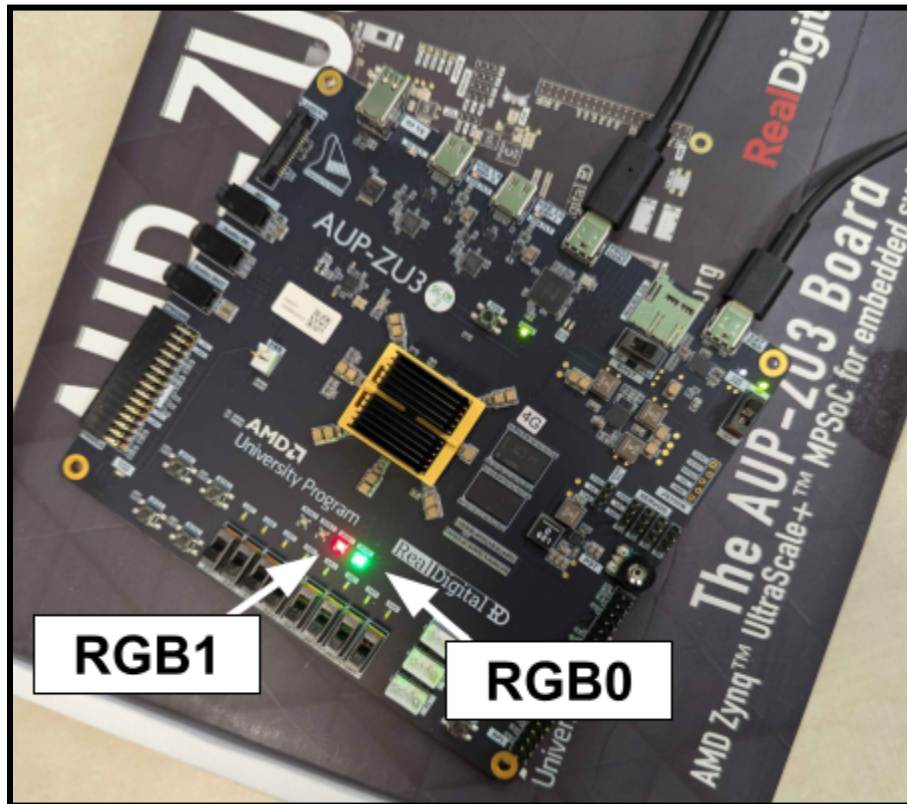


2.3 - Progress to Date

The main deliverable completed is the loading of the benign traffic light program, written in VHDL, onto the FPGA; the program is a simple finite-state machine that performs the following:

- Both LEDs labeled RGB0 and RGB1 are initially red for 3 seconds.
- RGB0 turns green for 20 seconds.
- RGB0 turns yellow for 3 seconds.
- Both RGB0 and RGB1 are red for 3 seconds.
- RGB1 turns green for 20 seconds.

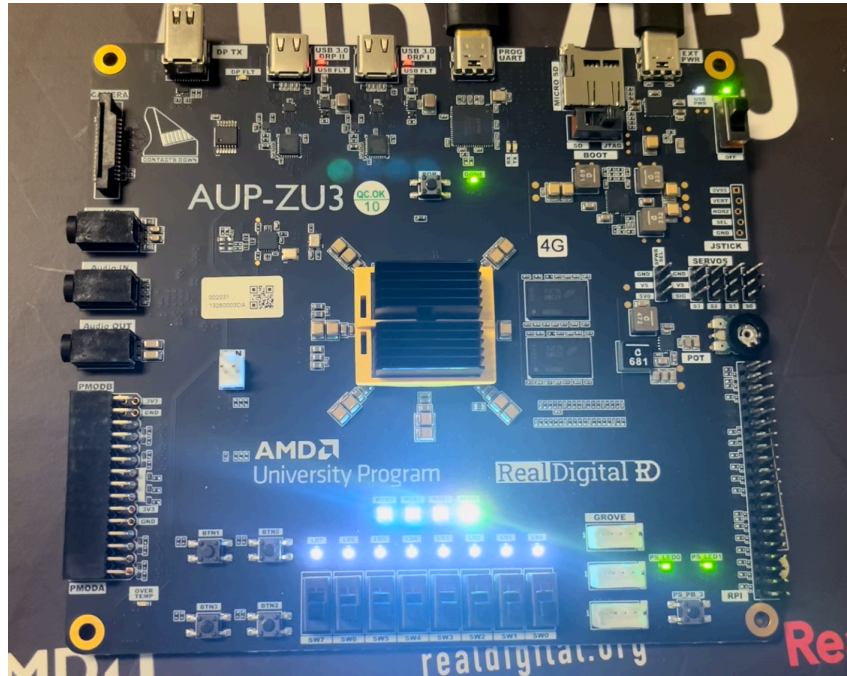
- RGB1 turns yellow for 3 seconds.
- (the cycle repeats.)



The FPGA we are using (RealDigital's AUP-ZU3 board), with the traffic light program running.

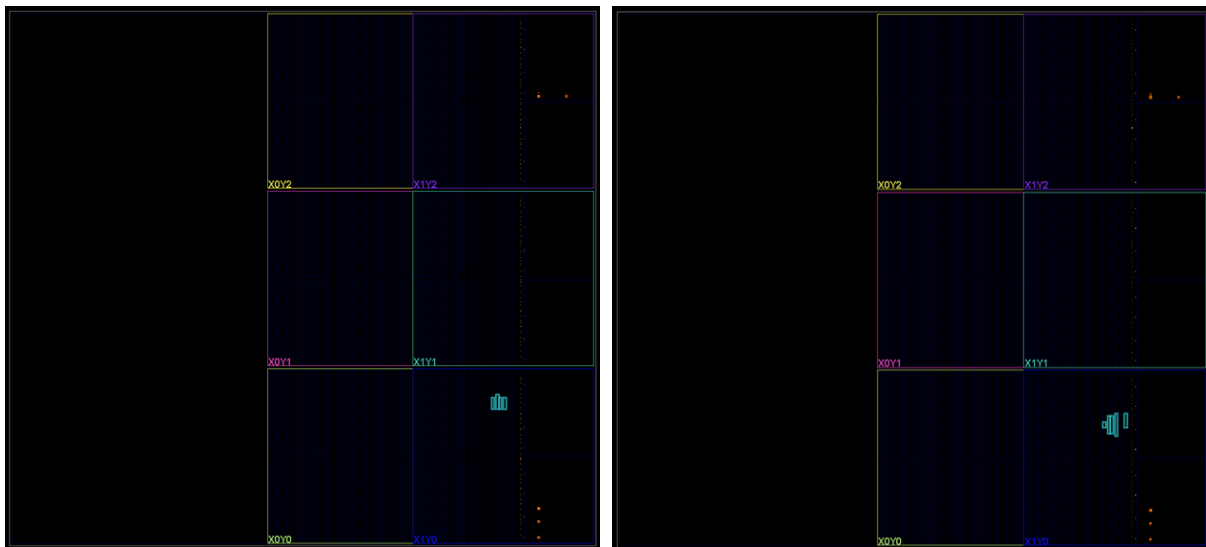
Additionally, Verilog and Vitis HLS (high-level synthesis) versions of the traffic light program were created and loaded onto the FPGA. The HLS version was written in C++, then synthesized into Verilog. Alterations to the generated Verilog code were only made to properly connect input and output pins to the AUP-ZU3 board's clock and LEDs. This program is a finite state machine that behaves similar to the VHDL design and performs the following:

- RGB0 turns green for 20 seconds
- RGB0 turns yellow for 3 seconds
- RGB0 turns red for 20 seconds
- (the cycle repeats)



The AUP-ZU3 board, with the HLS traffic light program running in the “yellow” state

It was observed that the hardware implementation for the low-level Verilog version and the HLS C++ version differ, despite the fact that both implementations were synthesized and implemented from Verilog. The team intends to investigate if differences can also be found through side-channel analysis.

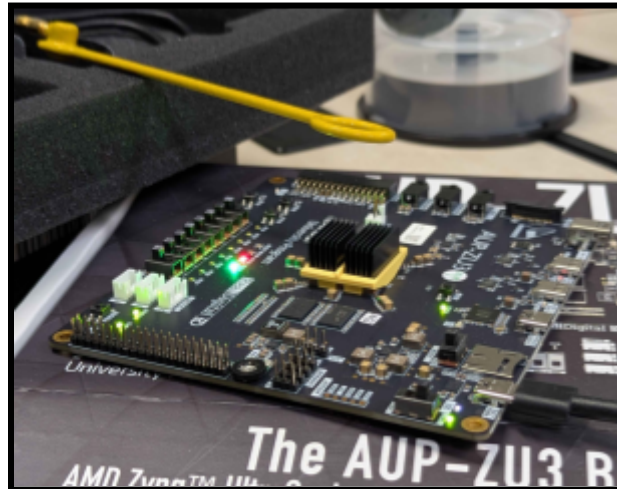


Hardware implementations for the low-level Verilog traffic light program (left) and the HLS program (right)

As additional benign programs that are more complex than a simple traffic light, an elevator controller has been written in VHDL and successfully loaded onto the FPGA, and a vending machine is being written in C++ for Vitis HLS and Verilog. Since the traffic light program is a

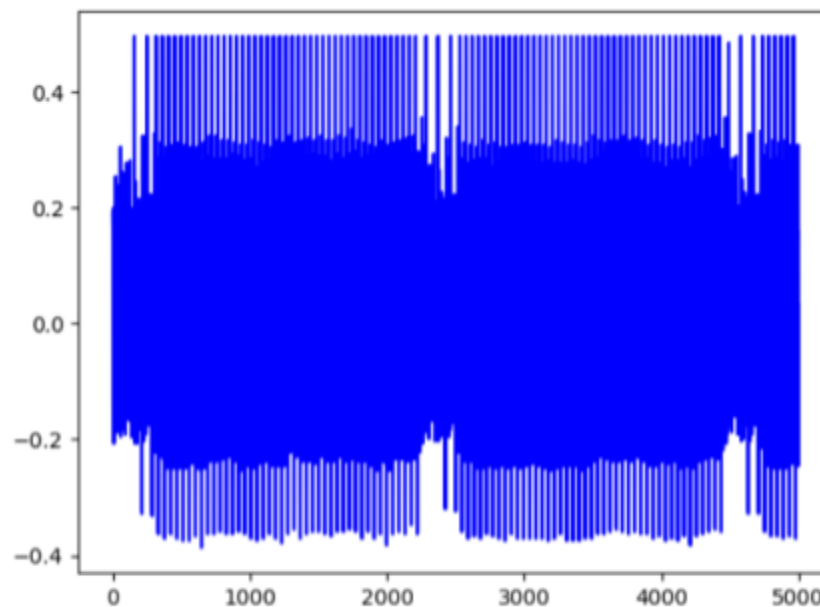
simple finite state machine, the team is intending to develop more complex programs to produce stronger readings for side-channel analysis.

Side-channel measurements and analysis are currently in the very early stages since we only recently loaded a program onto the board. Preliminary setups for electromagnetic emission SCA and power SCA have been tested, although a final experimental setup is yet to be determined.



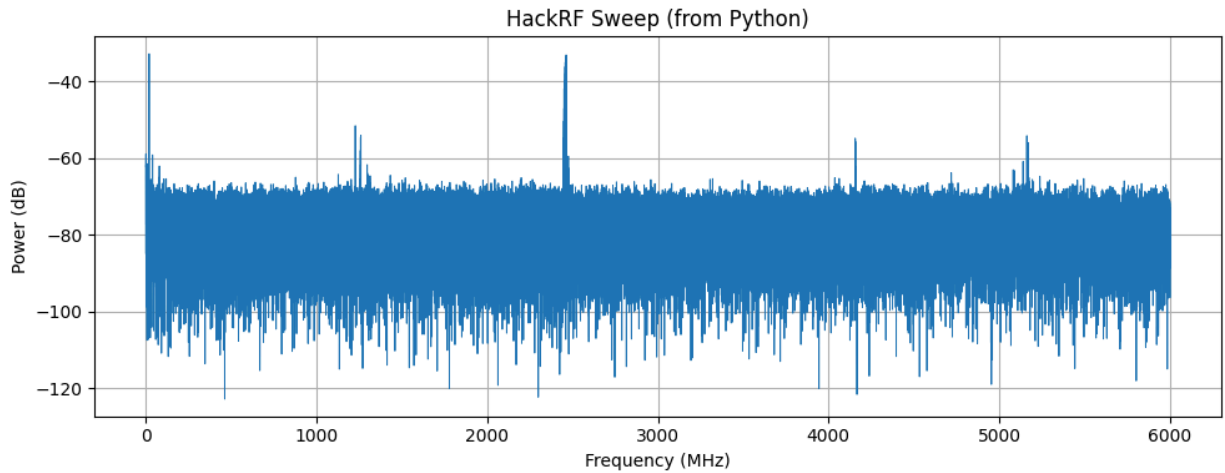
Our very advanced HackRF setup for measuring electromagnetic emissions. We do not know the distance of measurement.

Power analysis tests using the ChipWhisperer Nano have been conducted using the on-board target. Power traces were collected from the target and plotted in Python using Matplotlib. This is representative of how power data will be collected from the FPGA, albeit with refinements.



Power trace collected in Jupyter Notepad with the ChipWhisperer Nano

Electromagnetic analysis tests using the HackRF One have been conducted against the FPGA running the above bitstreams. The first step of analysis was to conduct a frequency sweep to detect the strongest emissions among the processor's harmonics. After narrowing down the frequencies, we began to take measurements while adjusting parameters of the probe. As of now the strongest emission observed is around the 2000 to 3000 MHz range. Further analysis and noise reduction is necessary to record concrete data.



3.0 - Summary of Results

After learning two hardware description languages (VHDL and Verilog) and Vitis HLS software, our team has successfully developed a benign traffic light and elevator program and loaded them onto an FPGA using Vivado. We have synthesized and implemented low-level Verilog and HLS versions of the traffic light program and identified differences in hardware implementation. Preliminary exploration of side-channel analysis techniques and experimentation setups utilizing the HackRF and ChipWhisperer are ongoing.

We will utilize our knowledge and experiments to gather concrete experimental data on side-channel analysis of FPGA programs during our next steps for this project. With our test program already implemented and tested, we are in a good position to develop a hardware trojan for the program. We will develop testbeds for electromagnetic and power SCA, refining their designs using preliminary data collected from side-channel analysis. Final results will be drawn from electromagnetic emission and power readings from the running FPGA program, which will be interpreted by the team to determine if hardware trojans can indeed be detected, and if our methodology is effective for such detection.

4.0 - Issues and Challenges

Our primary challenge encountered so far is perhaps one of the most common ones encountered in computer science as a whole - troubleshooting. When we initially received the FPGAs, we had a few cables ready to go for powering the boards and connecting them to our computers, but Vivado did not recognize that we were connected to the boards. After a few hours of investigating this issue with no leads, we determined that A) not all of our power cables were data transfer-capable and B) our initial power supply was not strong enough to provide 9W @ 3A for the board to fully power on.

Other challenges we experienced are (re)learning hardware description languages for the sake of developing hardware programs, and dealing with delays in shipping for our hardware, particularly the AUP-ZU3 boards containing our FPGAs. The first challenge fortunately did not pose much of a problem, but the second challenge caused a stall in the team's progress as we had programs written before receiving the hardware, but had no means of fully debugging or visualizing them on real hardware. We tackled this problem by working on other tasks and having our point-of-contact communicate with the company about the order status.

Going forward, the challenges we expect to face during the side-channel analysis phase of the project include developing an effective testbed for running experiments, gathering data that is significant and readable, and analyzing the data to determine if hardware trojans can indeed be detected using side-channel analysis. Our electromagnetic SCA setup must be able to support the probe and hold it at the intended distance from the board. Power SCA would require interpreting schematics to properly connect the board's power rails to power measurement ports on the ChipWhisperer.

5.0 - Bibliography

R. M. Rad, X. Wang, M. Tehranipoor and J. Plusquellic, "Power supply signal calibration techniques for improving detection resolution to hardware Trojans," 2008 IEEE/ACM International Conference on Computer-Aided Design, San Jose, CA, USA, 2008, pp. 632-639, doi: 10.1109/ICCAD.2008.4681643.

R. S. Chakraborty, I. Saha, A. Palchaudhuri, and G. K. Naik, "Hardware Trojan Insertion by Direct Modification of FPGA Configuration Bitstream," IEEE Design & Test, vol. 30, no. 2, pp. 45–54, Apr. 2013, doi: <https://doi.org/10.1109/mdt.2013.2247460>.

Sayakkara, A., Le-Khac, N.-A., & Scanlon, M. (2019). Leveraging Electromagnetic Side-Channel Analysis for the Investigation of IoT Devices. Digital Investigation, 29, S94–S103. <https://doi.org/10.1016/j.diin.2019.04.012>

R. Callan, A. Zajic and M. Prvulovic, "A Practical Methodology for Measuring the Side-Channel Signal Available to the Attacker for Instruction-Level Events," 2014 47th Annual IEEE/ACM International Symposium on Microarchitecture, Cambridge, UK, 2014, pp. 242-254, doi: 10.1109/MICRO.2014.39.

F. Schellenberg, D. R. E. Gnad, A. Moradi, and M. B. Tahoori, "An inside job: Remote power analysis attacks on FPGAs," 2018 Design, Automation & Test in Europe Conference & Exhibition (DATE), Mar. 2018, doi: 10.23919/date.2018.8342177.

Appendix A

A.1 - Acronyms

- **EM** - Electromagnetic
- **EM-SCA** - Electromagnetic side-channel analysis
- **FPGA** - Field-programmable gate array
- **FSM** - Finite state machine
- **HLS** - High level synthesis
- **SCA** - Side-channel analysis
- **SDR** - Software-defined radio

A.2 - Definitions

- **Hardware Trojan:** A malicious modification to a circuit or logic that is disguised as being a part of the circuit, but has some harmful function.
- **FPGA:** AKA Field-Programmable Gate Array - is a configurable integrated circuit that can be continuously reprogrammed by users to perform some function.
- **Bitstream:** A file containing the data to be loaded onto the FPGA - this includes the main FPGA logic (program) and configuration data for routing, memory, and I/O.
- **Side-Channel Analysis:** Obtaining (potentially sensitive) information from a system through measuring the side effects of its main computations, such as power consumption and EM emissions.
- **Signal Processing:** Extracting information from a signal (sound, images, sensor data)
- **Supply Chain:** The life cycle of a hardware component, where at each point an attacker can add some malicious function